

Server Authentication

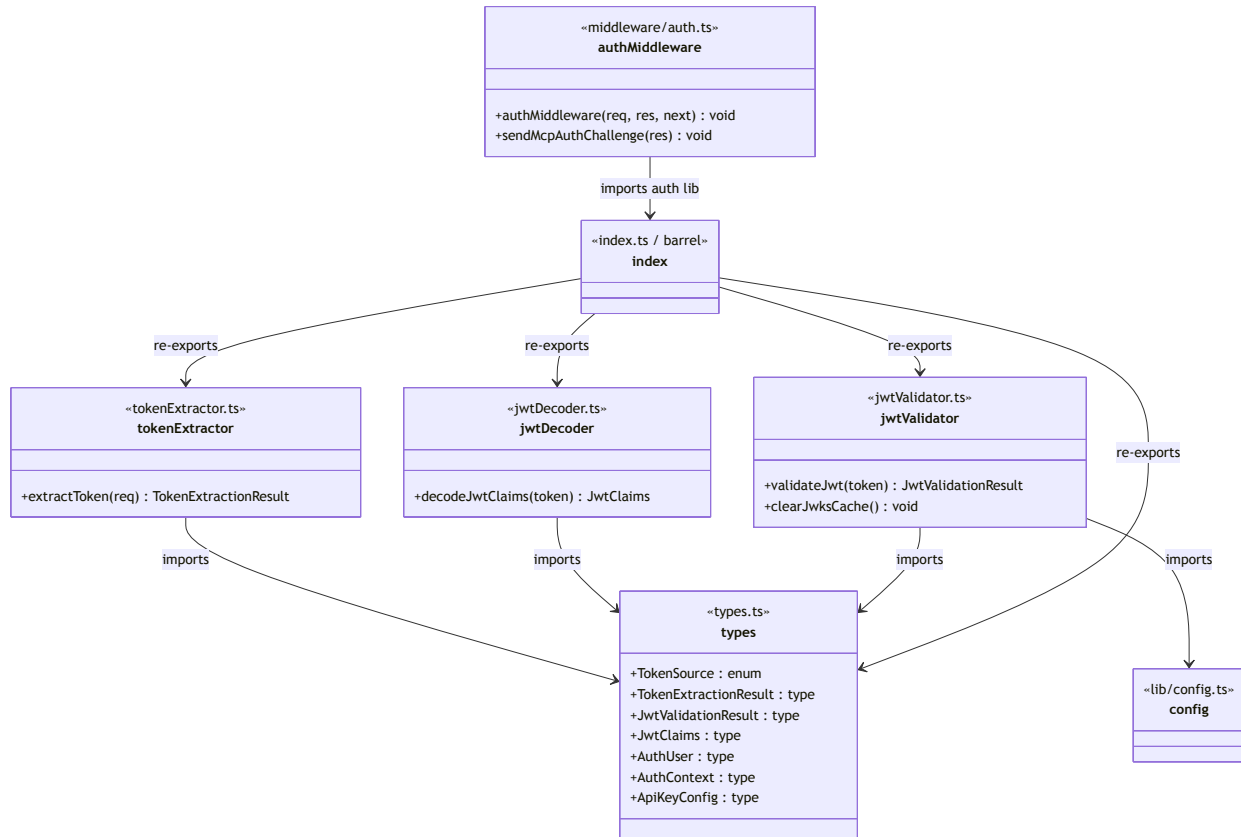
Overview

The Server Authentication module provides JWT-based authentication for the LiminalDB HTTP service layer. It is organized as a layered pipeline: token extraction from incoming requests, JWT decoding and cryptographic validation (via JWKS), and an Express-style auth middleware that gates access to all API routes. The module also supports MCP (Model Context Protocol) auth challenges for machine-client scenarios. All auth types are centralized in a shared `types.ts` file, and the barrel `index.ts` re-exports the full public surface.

Responsibilities

- Extract bearer tokens from HTTP request headers (Authorization header and other sources)
- Decode JWT claims without verification for inspection purposes
- Validate JWTs cryptographically using JWKS with caching (and cache-clearing support)
- Define shared auth types: `TokenSource`, `TokenExtractionResult`, `JwtValidationResult`, `JwtClaims`, `AuthUser`, `AuthContext`, `ApiKeyConfig`
- Provide Express auth middleware that authenticates requests and populates auth context
- Send MCP-compliant auth challenge responses for unauthenticated machine clients

Structure Diagram

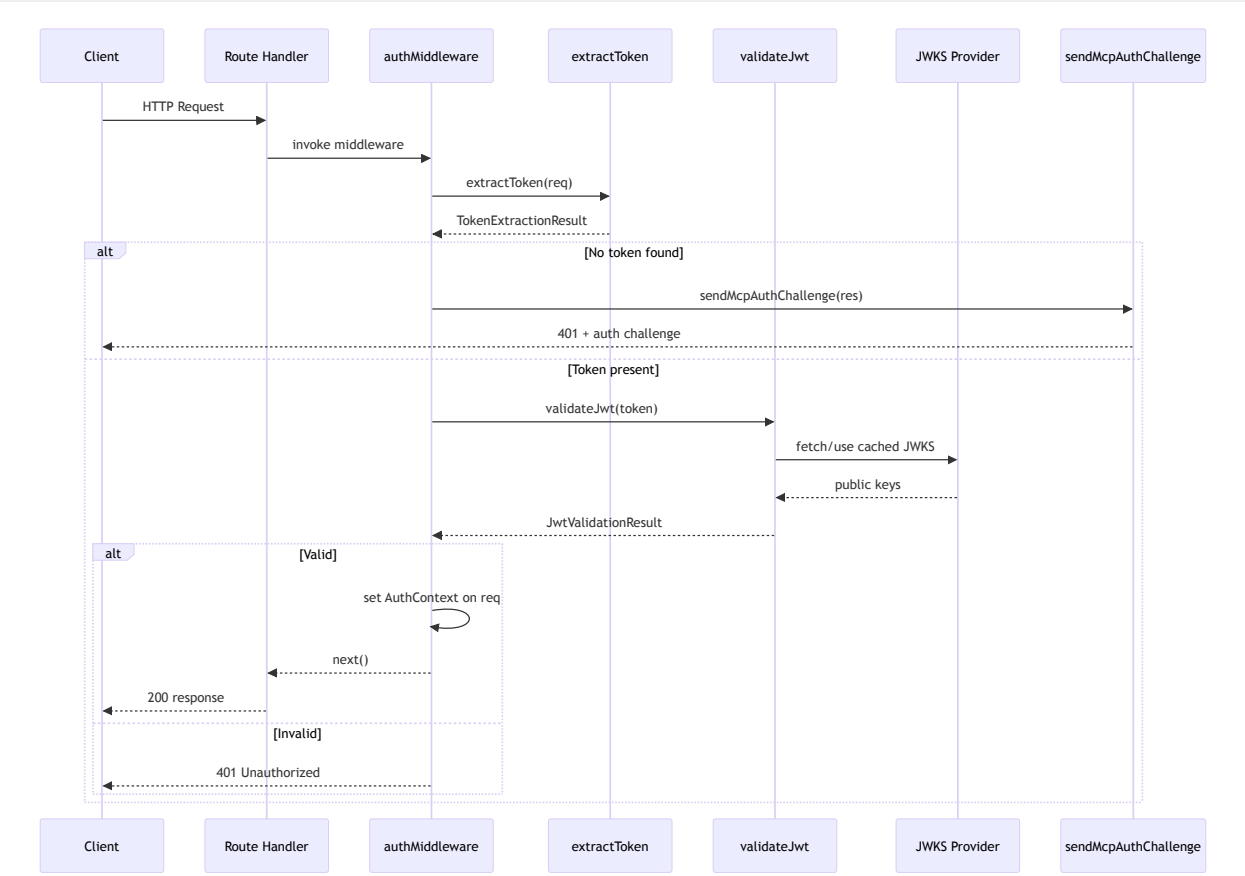


Entity Table

Name	Kind	Role
types.ts	type-definitions	Defines all shared auth types and enums (<code>TokenSource</code> , <code>TokenExtractionResult</code> , <code>JwtValidationResult</code> , <code>JwtClaims</code> , <code>AuthUser</code> , <code>AuthContext</code> , <code>ApiKeyConfig</code>)
extractToken	function	Extracts bearer token from incoming HTTP request, returning a <code>TokenExtractionResult</code> with the

Name	Kind	Role
decodeJwtClaims	function	Decodes JWT payload claims without cryptographic verification
validateJwt	function	Validates a JWT cryptographically using JWKS fetched from the configured provider, with resul
clearJwksCache	function	Clears the cached JWKS key set, forcing re-fetch on next validation
index.ts (barrel)	file	Barrel re-export for the auth library — aggregates all auth functions and types into a single imp
authMiddleware	function	Express middleware that extracts token, validates JWT, and populates AuthContext on the requ
sendMcpAuthChallenge	function	Sends an MCP-compliant authentication challenge response for machine clients that lack valid

Key Flow



Flow Notes

Step	Actor/Component	Action
1	Client	Sends HTTP request with Authorization header (Bearer token) to an API route
2	authMiddleware	Invokes extractToken to parse the bearer token from the request headers
3	authMiddleware	If no token is found, delegates to sendMcpAuthChallenge to return a 401 with an MCP-compliant challenge
4	authMiddleware	If token is present, calls validateJwt which verifies the JWT signature against JWKS keys (cached or freshly fetc
5	authMiddleware	On successful validation, populates AuthContext on the request object and calls next()

Source Coverage

- src/lib/auth/index.ts
- src/lib/auth/jwtDecoder.ts
- src/lib/auth/jwtValidator.ts
- src/lib/auth/tokenExtractor.ts
- src/lib/auth/types.ts
- src/middleware/auth.ts

Cross-Module Context

- src/api/health.ts -> src/middleware/auth.ts (import)
- src/api/mcp.ts -> src/lib/auth/index.ts (import)
- src/api/mcp.ts -> src/middleware/auth.ts (import)
- src/index.ts -> src/lib/auth/tokenExtractor.ts (usage)
- src/index.ts -> src/middleware/auth.ts (usage)
- src/lib/auth/jwtValidator.ts -> src/lib/config.ts (import)
- src/routes/app.ts -> src/middleware/auth.ts (import)
- src/routes/auth.ts -> src/middleware/auth.ts (import)
- src/routes/drafts.ts -> src/middleware/auth.ts (import)
- src/routes/import-export.ts -> src/middleware/auth.ts (import)
- src/routes/preferences.ts -> src/middleware/auth.ts (import)
- src/routes/prompts.ts -> src/middleware/auth.ts (import)
- tests/service/auth/mcp.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/auth/mcp.test.ts -> src/middleware/auth.ts (usage)
- tests/service/auth/middleware.test.ts -> src/lib/auth/index.ts (usage)
- tests/service/auth/middleware.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/auth/middleware.test.ts -> src/middleware/auth.ts (usage)
- tests/service/auth/routes.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/auth/routes.test.ts -> src/middleware/auth.ts (usage)
- tests/service/drafts/drafts.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/drafts/drafts.test.ts -> src/middleware/auth.ts (usage)
- tests/service/mcp/auth-challenge.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/mcp/auth-challenge.test.ts -> src/middleware/auth.ts (usage)
- tests/service/mcp/resources.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/mcp/resources.test.ts -> src/middleware/auth.ts (usage)
- tests/service/mcp/tools.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/mcp/tools.test.ts -> src/middleware/auth.ts (usage)
- tests/service/mcp/well-known.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/mcp/well-known.test.ts -> src/middleware/auth.ts (usage)
- tests/service/preferences.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/preferences.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/createPrompts.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/createPrompts.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/deletePrompt.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/deletePrompt.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/edgeCases.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/edgeCases.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/flagsUsage.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/flagsUsage.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/getPrompt.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/getPrompt.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/importExport.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/importExport.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/listPrompts.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/listPrompts.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/mcpTools.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/mcpTools.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/mergePrompt.test.ts -> src/lib/auth/tokenExtractor.ts (usage)

- tests/service/prompts/mergePrompt.test.ts -> src/middleware/auth.ts (usage)
- tests/service/prompts/updatePrompt.test.ts -> src/lib/auth/tokenExtractor.ts (usage)
- tests/service/prompts/updatePrompt.test.ts -> src/middleware/auth.ts (usage)